Advanced Network Security and Architectures

METI

Project Report [EN]

Authors:

José Oliveira (103771) jose.novais.oliveira@tecnico.ulisboa.pt

2025/2026 – 2nd Semester, P4

Contents

1 Introduction 2

2 Topology and Base Configuration 2

3 GETVPN Configuration 3
- 3.1 Overview of the Configuration Model . . . 3
- 3.2 Key Server Configuration (R1) . . . 4
  - 3.2.1 IKE Phase 1 Policy . . . 4
  - 3.2.2 Transform Set and IPSec Profile . . . 4
  - 3.2.3 RSA Key Pair . . . 4
  - 3.2.4 Traffic Selector ACL . . . 4
  - 3.2.5 GDOI Group . . . 5
- 3.3 Group Member Configuration (R2, R3, R4) . . . 5

4 Verification 6
- 4.1 Key Server Status . . . 6
- 4.2 Group Member Status . . . 9

5 Lab Tasks 11
- 5.1 IKE and GROUPKEY-PULL Analysis . . . 11
  - 5.1.1 IKE Phase 1 – Main Mode . . . 12
  - 5.1.2 GROUPKEY-PULL . . . 13
  - 5.1.3 Repeated Pattern Across All Group Members . . . 15
- 5.2 SPI Consistency Across Group Members . . . 15
  - 5.2.1 Wireshark Capture . . . 16
  - 5.2.2 CLI Verification . . . 17
  - 5.2.3 Why All SPIs Are Identical . . . 18
- 5.3 TEK Rekey Observation . . . 18

6 GETVPN Extensions 18

7 Conclusion 18

1 Introduction

Group Encrypted Transport VPN (GETVPN) is a Cisco proprietary technology designed to provide scalable, any-to-any encrypted communication over private WAN infrastructures such as MPLS networks. Unlike traditional point-to-point IPSec VPNs, which require a dedicated tunnel between each pair of sites, GETVPN operates on a group key model: all participating routers share the same encryption keys and policy, allowing any member to securely communicate with any other member without the overhead of per-pair tunnel establishment.

The GETVPN architecture is built around two roles. The **Key Server (KS)** is responsible for generating and distributing cryptographic keys and security policies to all group participants. The **Group Members (GMs)** are the routers that register with the Key Server, receive the shared keys and traffic policies, and then encrypt or decrypt data traffic autonomously. Once registered, GMs communicate directly with each other; the Key Server is not involved in the data plane.

The goal of this project is to deploy and study a GETVPN solution in a simulated environment using GNS3, analyse the key management protocol exchanges, observe traffic encryption behaviour, and extend the base configuration with additional features. The remainder of this report describes the topology, the configuration steps, the observations made during the lab tasks, and the extensions implemented.

2 Topology and Base Configuration

The network topology used in this project is shown in Figure 1. It consists of four Cisco IOSv routers interconnected through a central switch. Router R1 acts as the Key Server and is connected to the shared segment `192.168.1.0/24` with address `192.168.1.254`. Routers R2, R3, and R4 are the Group Members, with addresses `192.168.1.2`, `192.168.1.3`, and `192.168.1.4` respectively on their `gi0/0` interfaces. Each Group Member also has a LAN-facing interface (`gi0/1`) connecting to a dedicated subnet: R2 serves `192.168.10.0/24`, R3 serves `192.168.20.0/24`, and R4 serves `192.168.30.0/24`.

Three end hosts (PC1, PC2, PC3) are attached to these LAN subnets. Since the lab environment does not include dedicated PC images, Cisco 3725 routers were used to simulate the hosts; IP routing was disabled on these devices (`no ip routing`), and a default gateway static route was configured on each to point to the respective Group Member. Each simulated PC was assigned the `.100` address of its subnet (e.g., PC1 at `192.168.10.100`) and its icon was changed to a PC in GNS3 for clarity.

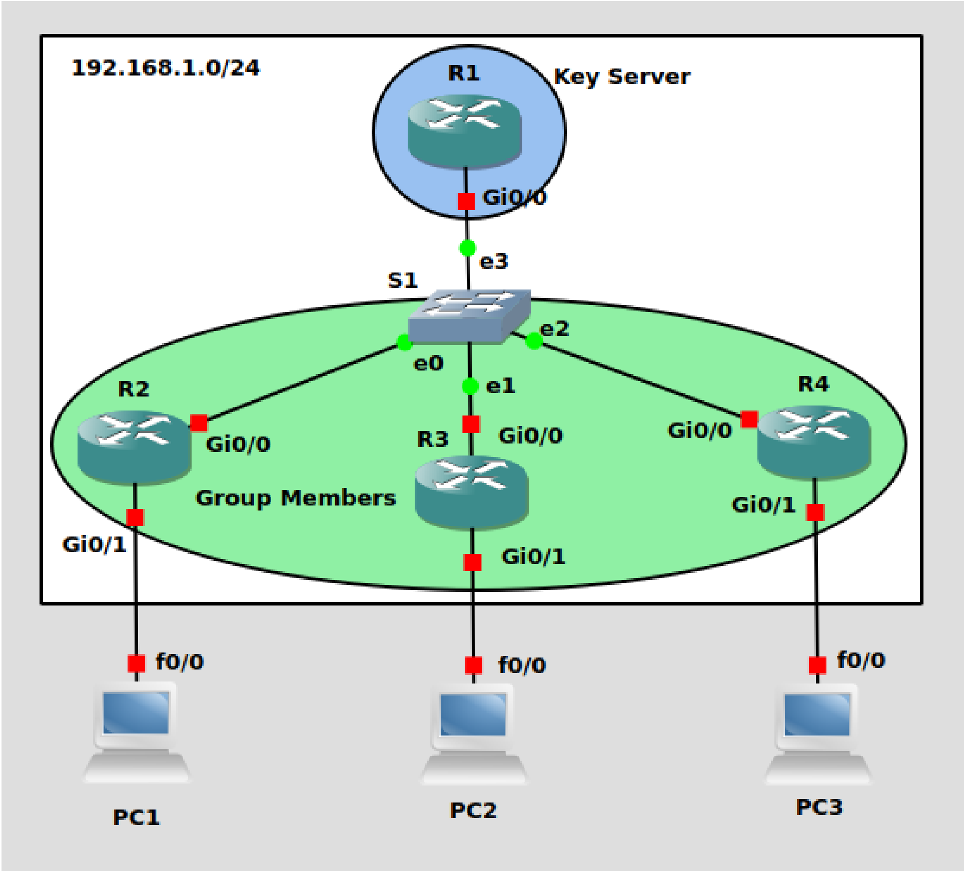


Figure 1: Project topology. R1 is the Key Server; R2, R3, and R4 are Group Members.

OSPF was chosen as the routing protocol and was configured on all four routers to advertise all connected networks. Prior to applying any GETVPN or IPSec configuration, end-to-end reachability was verified by pinging between all three PCs and between all router interfaces. All pings were successful, confirming that the routing layer was fully operational before any security configuration was introduced.

3 GETVPN Configuration

3.1 Overview of the Configuration Model

GETVPN security configuration is divided between the Key Server and the Group Members. The Key Server defines all security policy — the IKE parameters, the cryptographic algorithms, the traffic selector (which packets to encrypt), and the keys themselves — and distributes this policy to GMs during their registration. GMs only need to know how to authenticate with the KS and where to register; they receive everything else from the KS automatically.

The key management protocol used to distribute keys and policy is **GDOI (Group Domain of Interpretation)**, which runs over UDP port 848. The registration process has two phases: an IKE Phase 1 exchange that establishes a secure authenticated channel between the GM and the KS, followed by a **GROUPKEY-PULL** exchange in which the GM requests and receives the group keys and IPSec policy.

3.2 Key Server Configuration (R1)

3.2.1 IKE Phase 1 Policy

The first step is to define the parameters for the IKE Phase 1 exchange that will protect the GDOI registration channel between each GM and the KS. AES encryption, SHA hashing, pre-shared key authentication, and Diffie-Hellman group 5 are used. The wildcard address `0.0.0.0` allows the KS to accept registrations from any GM without listing each one individually.

```
! IKE Phase 1
R1(config)#crypto isakmp policy 10
R1(config-isakmp)#encryption aes
R1(config-isakmp)#hash sha
R1(config-isakmp)#authentication pre-share
R1(config-isakmp)#group 5
R1(config)#crypto isakmp key saar address 0.0.0.0
```

3.2.2 Transform Set and IPSec Profile

The *transform set* defines the cryptographic algorithms that will be applied to data traffic: ESP with AES for encryption and SHA-HMAC for integrity. The *IPSec profile* bundles the transform set into a named object that is later referenced inside the GDOI group configuration.

```
! Transform set (cipher suite for data traffic)
R1(config)#crypto ipsec transform-set myTSet esp-aes esp-sha-hmac

! IPSec profile (references the transform set)
R1(config)#crypto ipsec profile myIPSecProfile
R1(ipsec-profile)#set transform-set myTSet
```

3.2.3 RSA Key Pair

The KS signs its rekey messages using an RSA key pair so that GMs can verify the authenticity and integrity of key updates. The label `myRSAKeys` is used to reference this key pair in the GDOI group configuration.

```
R1(config)#crypto key generate rsa modulus 1024 label myRSAKeys
```

3.2.4 Traffic Selector ACL

GETVPN does not encrypt all traffic indiscriminately. An extended ACL defines the *interesting traffic* — the flows that should be protected. This ACL is configured on the KS and pushed to all GMs during registration. In this lab, only ICMP traffic is protected, which is sufficient for the ping-based verification tasks.

```
! Interesting traffic (what gets encrypted)
R1(config)#ip access-list extended ICMP
R1(config-ext-nacl)#permit icmp any any
```

3.2.5 GDOI Group

The GDOI group is the central object of the KS configuration. It ties together the identity number (which GMs must match to join the group), the KS address, the rekey mechanism, and the IPSec SA policy including the traffic selector.

```
R1(config)#crypto gdoi group myGDOIGroup
R1(config-gkm-group)#identity number 123
R1(config-gkm-group)#server local
R1(gkm-local-server)#address ipv4 192.168.1.254
R1(gkm-local-server)#rekey authentication mypubkey rsa myRSAKeys
R1(gkm-local-server)#rekey transport unicast
R1(gkm-local-server)#sa ipsec 10
R1(gkm-sa-ipsec)#profile myIPSecProfile
R1(gkm-sa-ipsec)#match address ipv4 ICMP
```

The `rekey transport unicast` directive instructs the KS to send key refresh messages directly to each registered GM individually, as opposed to multicast delivery.

3.3 Group Member Configuration (R2, R3, R4)

The configuration is identical on all three GMs; only the hostname differs. Each GM configures the same IKE Phase 1 parameters and pre-shared key as the KS, joins the GDOI group by its identity number and the KS address, and applies a crypto map to its WAN-facing interface to activate the received policy.

```
! IKE Phase 1 (must match KS policy)
Rx(config)#crypto isakmp policy 10
Rx(config-isakmp)#encryption aes
Rx(config-isakmp)#hash sha
Rx(config-isakmp)#authentication pre-share
Rx(config-isakmp)#group 5
Rx(config)#crypto isakmp key saar address 192.168.1.254

! Transform set (required locally, actual policy comes from KS)
Rx(config)#crypto ipsec transform-set TRANSFORM_SET esp-aes esp-sha-hmac

! GDOI group (must match KS identity number)
Rx(config)#crypto gdoi group myGDOIGroup
Rx(config-gkm-group)#identity number 123
Rx(config-gkm-group)#server address ipv4 192.168.1.254

! Crypto map applied to WAN interface
Rx(config)#crypto map myCryptoMap 10 gdoi
Rx(config-crypto-map)#set group myGDOIGroup
Rx(config)#interface gi0/0
Rx(config-if)#crypto map myCryptoMap
```

It is worth noting that the transform set defined locally on a GM is not used for data traffic; the actual IPSec policy (including transform set, traffic selector, and keys) is always received from the KS. The local transform set is only relevant for the IKE Phase 1 exchange during registration.

4 Verification

4.1 Key Server Status

After all GMs were configured, the `show crypto gdoi` and `show crypto gdoi ks members` commands were used on R1 to confirm correct operation.

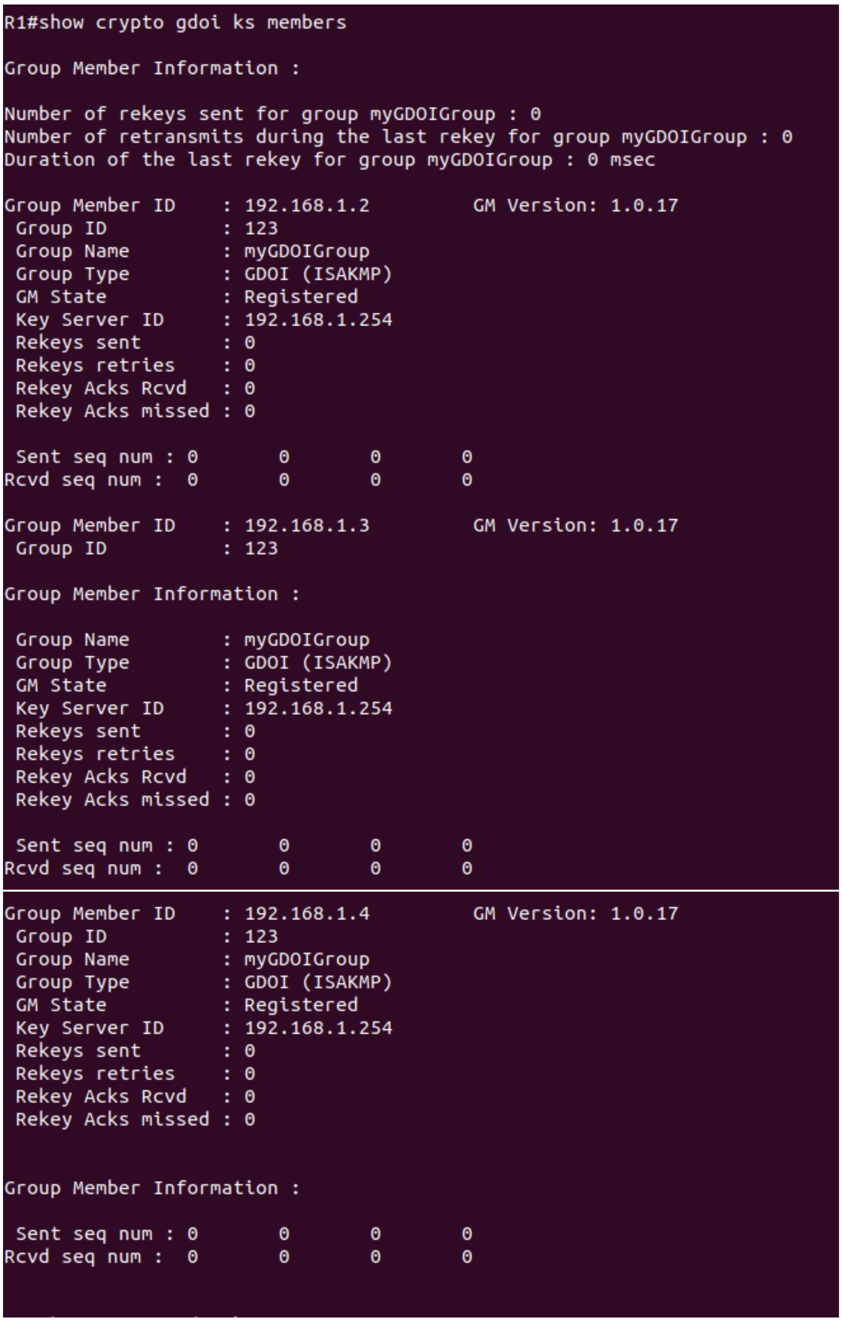
```
R1#show crypto gdoi
GROUP INFORMATION

    Group Name                : myGDOIGroup (Unicast)
    Re-auth on new CRL        : Disabled
    Group Identity            : 123
    Group Type                : GDOI (ISAKMP)
    Crypto Path               : ipv4
    Key Management Path       : ipv4
    Group Members             : 3
    IPSec SA Direction        : Both
    IP D3P Window             : Disabled
    CKM status                : Disabled
    Group Rekey Lifetime      : 86400 secs
    Group Rekey
        Remaining Lifetime    : 86355 secs
        Time to Rekey         : 86130 secs
        Acknowledgement Cfg   : Cisco
    Rekey Retransmit Period   : 10 secs
    Rekey Retransmit Attempts: 2
    Group Retransmit
        Remaining Lifetime    : 0 secs

      IPSec SA Number         : 10
      IPSec SA Rekey Lifetime: 3600 secs
      Profile Name            : myIPSecProfile
      Replay method           : Count Based
      Replay Window Size      : 64
      Tagging method          : Disabled
      SA Rekey
         Remaining Lifetime   : 3556 secs
        Time to Rekey         : 3170 secs
      ACL Configured          : access-list ICMP

     Group Server list        : Local
```

Figure 2: `show crypto gdoi` on the Key Server (R1). Three Group Members registered.



```
R1#show crypto gdoi ks members

Group Member Information :

Number of rekeys sent for group myGDOIGroup : 0
Number of retransmits during the last rekey for group myGDOIGroup : 0
Duration of the last rekey for group myGDOIGroup : 0 msec

Group Member ID    : 192.168.1.2        GM Version: 1.0.17
 Group ID          : 123
 Group Name        : myGDOIGroup
 Group Type        : GDOI (ISAKMP)
 GM State          : Registered
 Key Server ID     : 192.168.1.254
 Rekeys sent       : 0
 Rekeys retries    : 0
 Rekey Acks Rcvd   : 0
 Rekey Acks missed : 0

 Sent seq num : 0       0       0       0
Rcvd seq num :  0       0       0       0

Group Member ID    : 192.168.1.3        GM Version: 1.0.17
 Group ID          : 123

Group Member Information :

 Group Name        : myGDOIGroup
 Group Type        : GDOI (ISAKMP)
 GM State          : Registered
 Key Server ID     : 192.168.1.254
 Rekeys sent       : 0
 Rekeys retries    : 0
 Rekey Acks Rcvd   : 0
 Rekey Acks missed : 0

 Sent seq num : 0       0       0       0
Rcvd seq num :  0       0       0       0

Group Member ID    : 192.168.1.4        GM Version: 1.0.17
 Group ID          : 123
 Group Name        : myGDOIGroup
 Group Type        : GDOI (ISAKMP)
 GM State          : Registered
 Key Server ID     : 192.168.1.254
 Rekeys sent       : 0
 Rekeys retries    : 0
 Rekey Acks Rcvd   : 0
 Rekey Acks missed : 0


Group Member Information :

 Sent seq num : 0       0       0       0
Rcvd seq num :  0       0       0       0
```

Figure 3: `show crypto gdoi ks members` on R1. All three GMs (`192.168.1.2`, `.3`, `.4`) show `GM State: Registered`.

The output confirms that all three GMs successfully completed the GROUPKEY-PULL exchange. Each entry shows `GM State: Registered` and identifies the correct Key Server address (`192.168.1.254`). The rekey counters are all zero, which is expected since no TEK refresh has occurred yet.

4.2 Group Member Status

The `show crypto gdoi` and `show crypto gdoi ipsec sa` commands were run on R2 as a representative Group Member.

Several important aspects of GETVPN operation are visible in the output from Figure 4:

- **Registration status**: R2 shows `Registration status: Registered`, confirming it successfully joined the group.
- **ACL received from KS**: The traffic selector (`permit icmp any any`) was pushed by the KS and installed on R2 as `gdoi_group_myGDOIGroup_temp_acl`. GMs do not define this locally.
- **TEK policy**: The Traffic Encryption Key policy shows `transform:esp-aes esp-sha-hmac` and SPI `0x6D2F03B9`, with a remaining lifetime of 3478 seconds. This is the key used to encrypt ICMP traffic between GMs.
- **KEK policy**: The Key Encryption Key uses 3DES with a lifetime of 86276 seconds and is used to protect future rekey messages from the KS. Notably, the KEK uses 3DES regardless of the AES algorithm specified in the IKE policy; this is a known behaviour in the IOS version used.
- **Active TEK Number: 1**: Confirms one TEK is active and ready to encrypt matching traffic.

```
R2#show crypto gdoi
GROUP INFORMATION

    Group Name               : myGDOIGroup
    Group Identity           : 123
    Group Type               : GDOI (ISAKMP)
    Crypto Path              : ipv4
    Key Management Path      : ipv4
    Rekeys received          : 0
    IPSec SA Direction       : Both

     Group Server list       : 192.168.1.254

Group Member Information For Group myGDOIGroup:
    IPSec SA Direction       : Both
    ACL Received From KS     : gdoi_group_myGDOIGroup_temp_acl

    Group member             : 192.168.1.2      vrf: None
       Local addr/port       : 192.168.1.2/848
       Remote addr/port      : 192.168.1.254/848
       fvrf/ivrf             : None/None
       Version               : 1.0.17
       Registration status   : Registered
       Registered with       : 192.168.1.254
       Re-registers in       : 3229 sec
       Succeeded registration: 1
       Attempted registration: 1
       Last rekey from       : 0.0.0.0
       Last rekey seq num    : 0
       Unicast rekey received: 0
       Rekey ACKs sent       : 0
       Rekey Received        : never
       DP Error Monitoring   : OFF
       IPSEC init reg executed    : 0
       IPSEC init reg postponed   : 0
       Active TEK Number     : 1
       SA Track (OID/status) : disabled

    Rekeys cumulative
       Total received        : 0
       After latest register : 0
       Rekey Acks sents      : 0

 ACL Downloaded From KS 192.168.1.254:
   access-list   permit icmp any any

KEK POLICY:
    Rekey Transport Type     : Unicast
    Lifetime (secs)          : 86276
    Encrypt Algorithm        : 3DES
    Key Size                 : 192
    Sig Hash Algorithm       : HMAC_AUTH_SHA
    Sig Key Length (bits)    : 1296

TEK POLICY for the current KS-Policy ACEs Downloaded:
  GigabitEthernet0/0:
    IPsec SA:
        spi: 0x6D2F03B9(1831797689)
        KGS: Disabled
        transform: esp-aes esp-sha-hmac
        sa timing:remaining key lifetime (sec): (3478)
        Anti-Replay(Counter Based) : 64
        tag method : disabled
        alg key size: 16 (bytes)
        sig key size: 20 (bytes)
        encaps: ENCAPS_TUNNEL
```

Figure 4: `show crypto gdoi` on R2. Registration confirmed; TEK and KEK policies downloaded from KS.

5 Lab Tasks

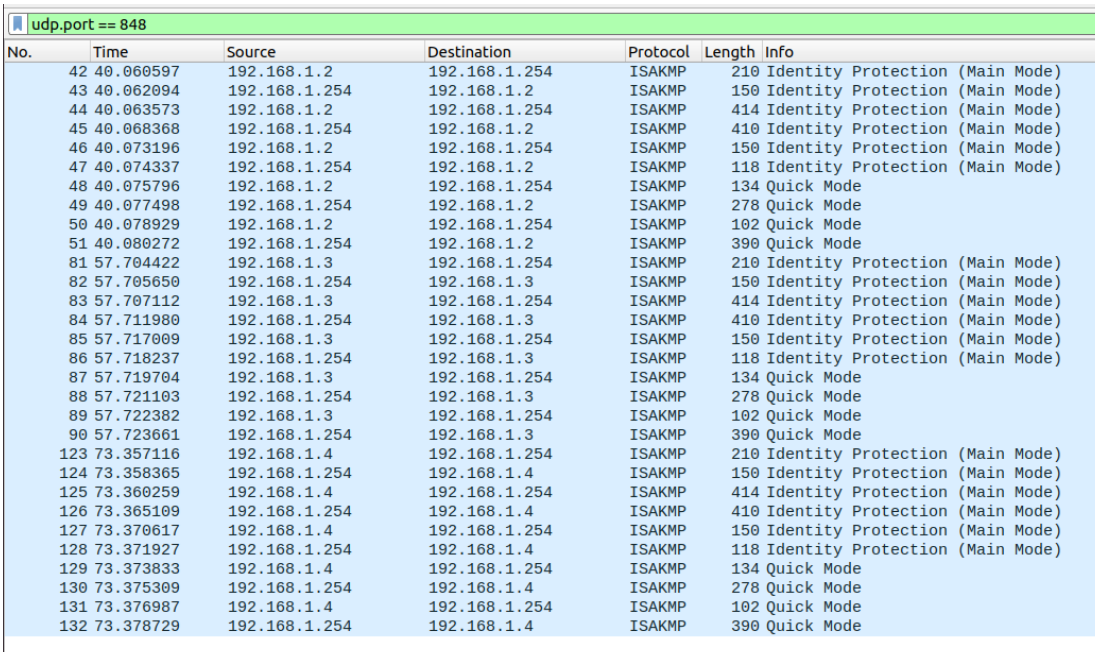
5.1 IKE and GROUPKEY-PULL Analysis

To capture the GDOI registration exchanges, a Wireshark probe was placed on the link between the Key Server (R1) and the central switch before any Group Member was started. R1 was brought up first, followed by R2, R3, and R4 in sequence. The traffic on this link runs on UDP port 848, which Wireshark does not decode automatically. Figure 5 shows the raw capture before applying any decoder, with packets appearing as generic UDP. The *Decode As* feature was used to instruct Wireshark to interpret port 848 traffic as ISAKMP, after which the exchange types became readable.

udp.port == 848

No.	Time	Source	Destination	Protocol	Length	Info
82	11.368100	192.168.1.2	192.168.1.254	UDP	210	848 → 848 Len=168
84	11.400215	192.168.1.254	192.168.1.2	UDP	150	848 → 848 Len=108
85	11.579027	192.168.1.2	192.168.1.254	UDP	414	848 → 848 Len=372
86	11.625902	192.168.1.254	192.168.1.2	UDP	410	848 → 848 Len=368
87	11.684127	192.168.1.3	192.168.1.254	UDP	210	848 → 848 Len=168
88	11.796866	192.168.1.2	192.168.1.254	UDP	150	848 → 848 Len=108
89	11.835710	192.168.1.254	192.168.1.2	UDP	118	848 → 848 Len=76
90	11.836983	192.168.1.254	192.168.1.3	UDP	150	848 → 848 Len=108
91	11.890377	192.168.1.3	192.168.1.254	UDP	414	848 → 848 Len=372
92	11.960476	192.168.1.4	192.168.1.254	UDP	210	848 → 848 Len=168
93	12.002840	192.168.1.2	192.168.1.254	UDP	134	848 → 848 Len=92
94	12.057683	192.168.1.254	192.168.1.3	UDP	410	848 → 848 Len=368
95	12.059283	192.168.1.254	192.168.1.4	UDP	150	848 → 848 Len=108
96	12.060734	192.168.1.254	192.168.1.2	UDP	278	848 → 848 Len=236
97	12.104609	192.168.1.3	192.168.1.254	UDP	150	848 → 848 Len=108
98	12.165482	192.168.1.4	192.168.1.254	UDP	414	848 → 848 Len=372
99	12.206568	192.168.1.2	192.168.1.254	UDP	102	848 → 848 Len=60
100	12.281874	192.168.1.254	192.168.1.3	UDP	118	848 → 848 Len=76
101	12.283891	192.168.1.254	192.168.1.2	UDP	390	848 → 848 Len=348
102	12.285297	192.168.1.254	192.168.1.4	UDP	410	848 → 848 Len=368
103	12.310353	192.168.1.3	192.168.1.254	UDP	134	848 → 848 Len=92
104	12.387496	192.168.1.4	192.168.1.254	UDP	150	848 → 848 Len=108
105	12.493902	192.168.1.254	192.168.1.4	UDP	118	848 → 848 Len=76
106	12.495173	192.168.1.254	192.168.1.3	UDP	278	848 → 848 Len=236
107	12.519393	192.168.1.3	192.168.1.254	UDP	102	848 → 848 Len=60
108	12.593413	192.168.1.4	192.168.1.254	UDP	134	848 → 848 Len=92
109	12.702747	192.168.1.254	192.168.1.3	UDP	390	848 → 848 Len=348
110	12.704298	192.168.1.254	192.168.1.4	UDP	278	848 → 848 Len=236
111	12.796863	192.168.1.4	192.168.1.254	UDP	102	848 → 848 Len=60
112	12.915493	192.168.1.254	192.168.1.4	UDP	390	848 → 848 Len=348

Figure 5: Capture filtered on UDP port 848 before applying the ISAKMP decoder.



udp.port == 848

No.	Time	Source	Destination	Protocol	Length	Info
42	40.060597	192.168.1.2	192.168.1.254	ISAKMP	210	Identity Protection (Main Mode)
43	40.062094	192.168.1.254	192.168.1.2	ISAKMP	150	Identity Protection (Main Mode)
44	40.063573	192.168.1.2	192.168.1.254	ISAKMP	414	Identity Protection (Main Mode)
45	40.068368	192.168.1.254	192.168.1.2	ISAKMP	410	Identity Protection (Main Mode)
46	40.073196	192.168.1.2	192.168.1.254	ISAKMP	150	Identity Protection (Main Mode)
47	40.074337	192.168.1.254	192.168.1.2	ISAKMP	118	Identity Protection (Main Mode)
48	40.075796	192.168.1.2	192.168.1.254	ISAKMP	134	Quick Mode
49	40.077498	192.168.1.254	192.168.1.2	ISAKMP	278	Quick Mode
50	40.078929	192.168.1.2	192.168.1.254	ISAKMP	102	Quick Mode
51	40.080272	192.168.1.254	192.168.1.2	ISAKMP	390	Quick Mode
81	57.704422	192.168.1.3	192.168.1.254	ISAKMP	210	Identity Protection (Main Mode)
82	57.705650	192.168.1.254	192.168.1.3	ISAKMP	150	Identity Protection (Main Mode)
83	57.707112	192.168.1.3	192.168.1.254	ISAKMP	414	Identity Protection (Main Mode)
84	57.711980	192.168.1.254	192.168.1.3	ISAKMP	410	Identity Protection (Main Mode)
85	57.717009	192.168.1.3	192.168.1.254	ISAKMP	150	Identity Protection (Main Mode)
86	57.718237	192.168.1.254	192.168.1.3	ISAKMP	118	Identity Protection (Main Mode)
87	57.719704	192.168.1.3	192.168.1.254	ISAKMP	134	Quick Mode
88	57.721103	192.168.1.254	192.168.1.3	ISAKMP	278	Quick Mode
89	57.722382	192.168.1.3	192.168.1.254	ISAKMP	102	Quick Mode
90	57.723661	192.168.1.254	192.168.1.3	ISAKMP	390	Quick Mode
123	73.357116	192.168.1.4	192.168.1.254	ISAKMP	210	Identity Protection (Main Mode)
124	73.358365	192.168.1.254	192.168.1.4	ISAKMP	150	Identity Protection (Main Mode)
125	73.360259	192.168.1.4	192.168.1.254	ISAKMP	414	Identity Protection (Main Mode)
126	73.365109	192.168.1.254	192.168.1.4	ISAKMP	410	Identity Protection (Main Mode)
127	73.370617	192.168.1.4	192.168.1.254	ISAKMP	150	Identity Protection (Main Mode)
128	73.371927	192.168.1.254	192.168.1.4	ISAKMP	118	Identity Protection (Main Mode)
129	73.373833	192.168.1.4	192.168.1.254	ISAKMP	134	Quick Mode
130	73.375309	192.168.1.254	192.168.1.4	ISAKMP	278	Quick Mode
131	73.376987	192.168.1.4	192.168.1.254	ISAKMP	102	Quick Mode
132	73.378729	192.168.1.254	192.168.1.4	ISAKMP	390	Quick Mode

Figure 6: Capture filtered on UDP port 848 after applying the ISAKMP decoder.

5.1.1 IKE Phase 1 – Main Mode

For each Group Member, the capture shows six ISAKMP messages labelled *Identity Protection (Main Mode)* (exchange type 2), constituting the IKE Phase 1 exchange. These six messages serve a single purpose: to establish a mutually authenticated, encrypted channel between the GM and the KS before any key material is exchanged. The three pairs of messages carry out the following steps: the first pair negotiates the SA parameters (encryption algorithm, hash, authentication method, and DH group); the second pair performs the Diffie-Hellman key exchange, from which both sides derive a shared secret; and the third pair carries the encrypted authentication payloads, where each side proves its identity using the pre-shared key `saar`.

A detail visible in the first message of each registration (Figure 7) is that the *Responder SPI* field is `0x0000000000000000`. This is expected: the KS has not yet responded, so no responder cookie has been assigned. By the end of Phase 1, both SPIs are populated, confirming the session is established. Also visible in this first message is the field *Domain of Interpretation: GDOI (2)*, which distinguishes this from a standard IPSec IKE negotiation (DOI 1) and confirms the session is being established specifically to support a GDOI group registration.

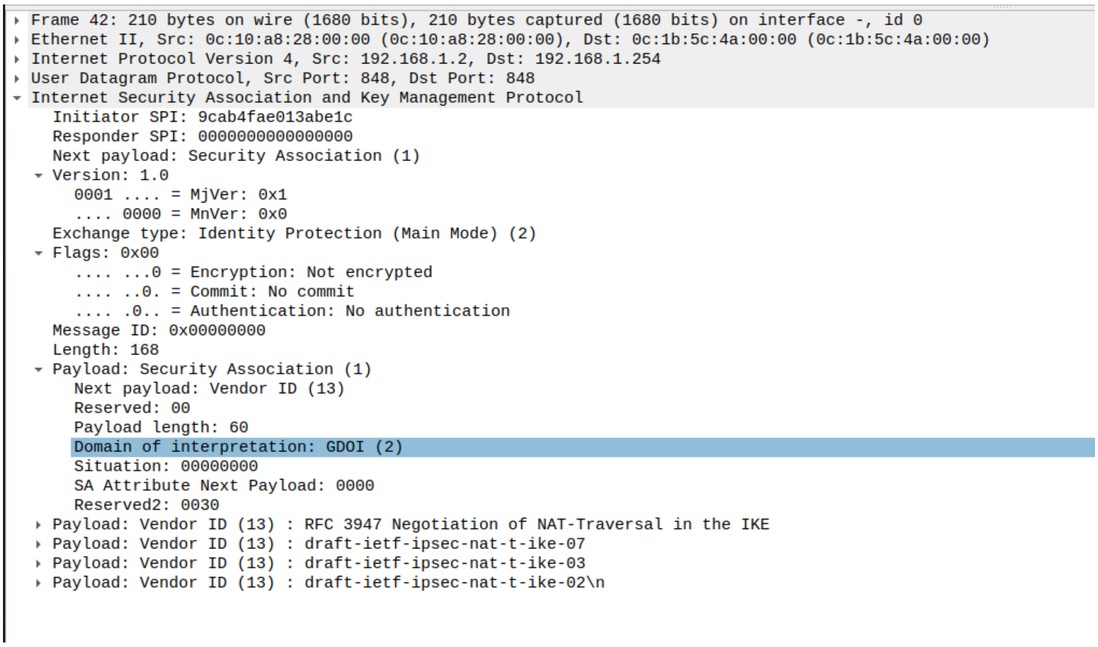


Figure 7: First IKE Phase 1 message expanded. Note the null Responder SPI and `Domain of Interpretation: GDOI (2)`.

5.1.2 GROUPKEY-PULL

Immediately following the six Phase 1 messages, four additional ISAKMP messages appear, labelled by Wireshark as *Quick Mode* (exchange type 32). These four messages constitute the **GROUPKEY-PULL** exchange, through which the GM requests and receives the group keys and IPSec policy from the KS. GDOI reuses the Quick Mode exchange type framing, which is why Wireshark labels them as such; the GDOI nature is confirmed by the DOI field observed in Phase 1.

Figures 8 and 8 show the four packets. All four share the same Message ID (`0x64b4a54f`), confirming they belong to the same exchange. The size of each packet is significant: the GM's initial request (packet 48) carries 64 bytes of encrypted data, while the KS responses carry 208 bytes (packet 49) and 320 bytes (packet 51). This asymmetry reflects the nature of the exchange — the GM sends a short registration request, while the KS response contains the full group policy: the Traffic Encryption Key (TEK), the Key Encryption Key (KEK), the traffic selector ACL, and the rekey parameters. All payloads are encrypted under the Phase 1 SA, so the key material itself is not visible in Wireshark. The fourth message (packet 50, 32 bytes) is the GM's acknowledgement completing the exchange.

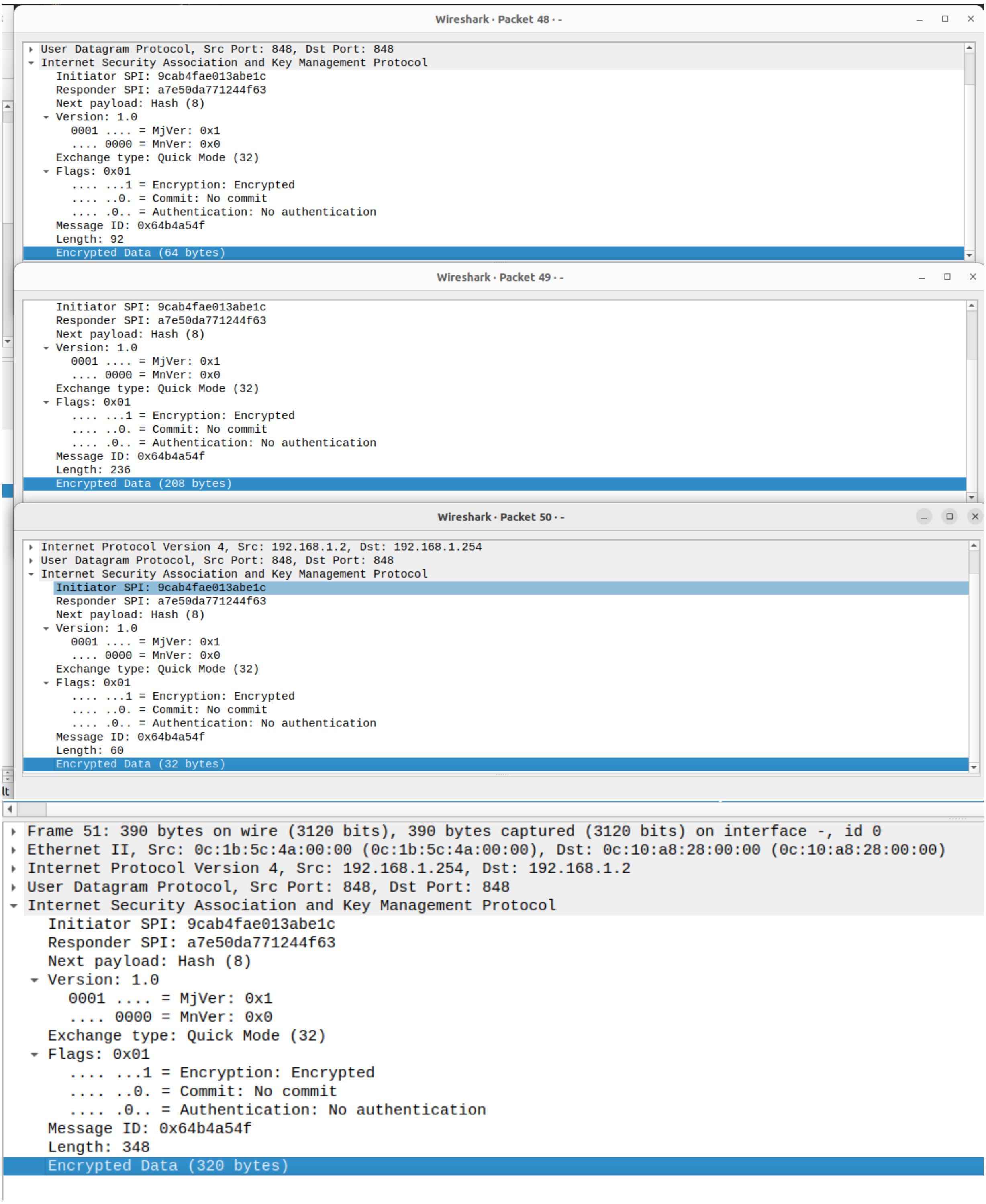
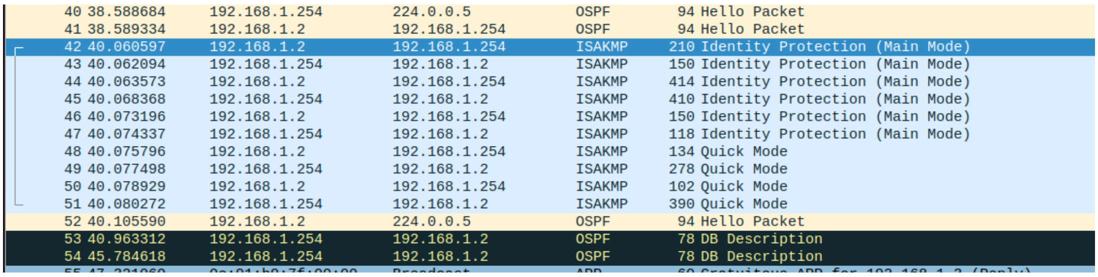


Figure 8: GROUPKEY-PULL exchange (Quick Mode, exchange type 32) between R2 and the Key Server.

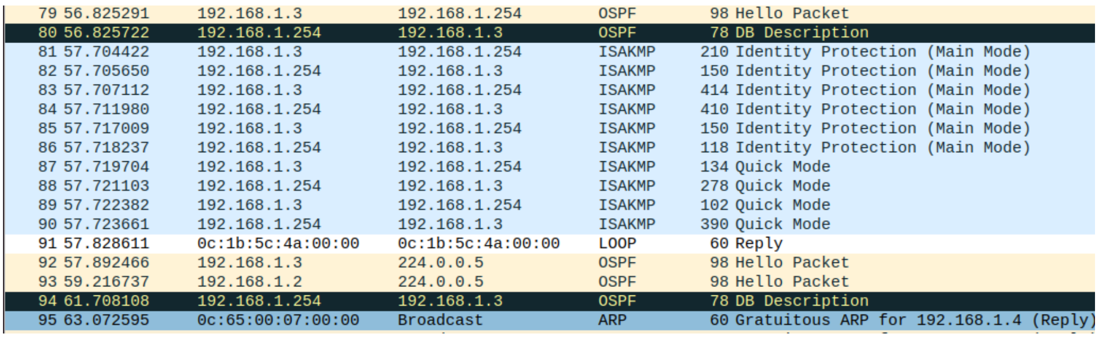
5.1.3 Repeated Pattern Across All Group Members

The identical 10-message pattern (6 Main Mode + 4 Quick Mode) was observed independently for each of the three Group Members, as shown in Figures 9, 10, and 11. Each GM performs its own IKE Phase 1 and GROUPKEY-PULL with the KS independently. The KS distributes the same TEK and policy to all three, which is what allows any GM to decrypt traffic encrypted by any other GM.



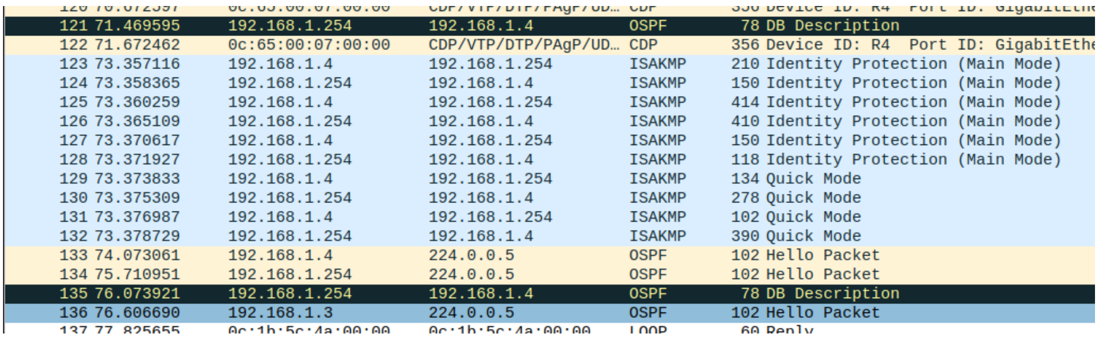
40	38.588684	192.168.1.254	224.0.0.5	OSPF	94 Hello Packet
41	38.589334	192.168.1.2	192.168.1.254	OSPF	94 Hello Packet
42	40.060597	192.168.1.2	192.168.1.254	ISAKMP	210 Identity Protection (Main Mode)
43	40.062094	192.168.1.254	192.168.1.2	ISAKMP	150 Identity Protection (Main Mode)
44	40.063573	192.168.1.2	192.168.1.254	ISAKMP	414 Identity Protection (Main Mode)
45	40.068368	192.168.1.254	192.168.1.2	ISAKMP	410 Identity Protection (Main Mode)
46	40.073196	192.168.1.2	192.168.1.254	ISAKMP	150 Identity Protection (Main Mode)
47	40.074337	192.168.1.254	192.168.1.2	ISAKMP	118 Identity Protection (Main Mode)
48	40.075796	192.168.1.2	192.168.1.254	ISAKMP	134 Quick Mode
49	40.077498	192.168.1.254	192.168.1.2	ISAKMP	278 Quick Mode
50	40.078929	192.168.1.2	192.168.1.254	ISAKMP	102 Quick Mode
51	40.080272	192.168.1.254	192.168.1.2	ISAKMP	390 Quick Mode
52	40.105590	192.168.1.2	224.0.0.5	OSPF	94 Hello Packet
53	40.963312	192.168.1.254	192.168.1.2	OSPF	78 DB Description
54	45.784618	192.168.1.254	192.168.1.2	OSPF	78 DB Description

Figure 9: 10-message registration sequence for R2 (packets 42–51).



79	56.825291	192.168.1.3	192.168.1.254	OSPF	98 Hello Packet
80	56.825722	192.168.1.254	192.168.1.3	OSPF	78 DB Description
81	57.704422	192.168.1.3	192.168.1.254	ISAKMP	210 Identity Protection (Main Mode)
82	57.705650	192.168.1.254	192.168.1.3	ISAKMP	150 Identity Protection (Main Mode)
83	57.707112	192.168.1.3	192.168.1.254	ISAKMP	414 Identity Protection (Main Mode)
84	57.711980	192.168.1.254	192.168.1.3	ISAKMP	410 Identity Protection (Main Mode)
85	57.717009	192.168.1.3	192.168.1.254	ISAKMP	150 Identity Protection (Main Mode)
86	57.718237	192.168.1.254	192.168.1.3	ISAKMP	118 Identity Protection (Main Mode)
87	57.719704	192.168.1.3	192.168.1.254	ISAKMP	134 Quick Mode
88	57.721103	192.168.1.254	192.168.1.3	ISAKMP	278 Quick Mode
89	57.722382	192.168.1.3	192.168.1.254	ISAKMP	102 Quick Mode
90	57.723661	192.168.1.254	192.168.1.3	ISAKMP	390 Quick Mode
91	57.828611	0c:1b:5c:4a:00:00	0c:1b:5c:4a:00:00	LOOP	60 Reply
92	57.892466	192.168.1.3	224.0.0.5	OSPF	98 Hello Packet
93	59.216737	192.168.1.2	224.0.0.5	OSPF	98 Hello Packet
94	61.708108	192.168.1.254	192.168.1.3	OSPF	78 DB Description
95	63.072595	0c:65:00:07:00:00	Broadcast	ARP	60 Gratuitous ARP for 192.168.1.4 (Reply)

Figure 10: 10-message registration sequence for R3.



121	71.469595	192.168.1.254	192.168.1.4	OSPF	78 DB Description
122	71.672462	0c:65:00:07:00:00	CDP/VTP/DTP/PAgP/UD…	CDP	356 Device ID: R4 Port ID: GigabitEthe
123	73.357116	192.168.1.4	192.168.1.254	ISAKMP	210 Identity Protection (Main Mode)
124	73.358365	192.168.1.254	192.168.1.4	ISAKMP	150 Identity Protection (Main Mode)
125	73.360259	192.168.1.4	192.168.1.254	ISAKMP	414 Identity Protection (Main Mode)
126	73.365109	192.168.1.254	192.168.1.4	ISAKMP	410 Identity Protection (Main Mode)
127	73.370617	192.168.1.4	192.168.1.254	ISAKMP	150 Identity Protection (Main Mode)
128	73.371927	192.168.1.254	192.168.1.4	ISAKMP	118 Identity Protection (Main Mode)
129	73.373833	192.168.1.4	192.168.1.254	ISAKMP	134 Quick Mode
130	73.375309	192.168.1.254	192.168.1.4	ISAKMP	278 Quick Mode
131	73.376987	192.168.1.4	192.168.1.254	ISAKMP	102 Quick Mode
132	73.378729	192.168.1.254	192.168.1.4	ISAKMP	390 Quick Mode
133	74.073061	192.168.1.4	224.0.0.5	OSPF	102 Hello Packet
134	75.710951	192.168.1.254	224.0.0.5	OSPF	102 Hello Packet
135	76.073921	192.168.1.254	192.168.1.4	OSPF	78 DB Description
136	76.606690	192.168.1.3	224.0.0.5	OSPF	102 Hello Packet

Figure 11: 10-message registration sequence for R4.

5.2 SPI Consistency Across Group Members

A fundamental property of GETVPN is that all Group Members share a single Traffic Encryption Key (TEK), meaning all encrypted traffic within the group carries the same Security

Parameter Index (SPI) regardless of source or destination. This was verified by capturing ESP traffic at the link between R2 and the switch while pinging from PC1 to both PC2 and PC3.

5.2.1 Wireshark Capture

Figures 12 and 13 show the ESP packets captured during the two ping sessions. In both cases — PC1 to PC2 (`192.168.10.100` → `192.168.20.100`) and PC1 to PC3 (`192.168.10.100` → `192.168.30.100`) — every ESP packet carries `SPI=0x24FF23B1`. Despite the traffic being destined for different Group Members, the SPI is identical in both captures. Expanding any individual packet confirms `ESP SPI: 0x24ff23b1 (620700593)`.

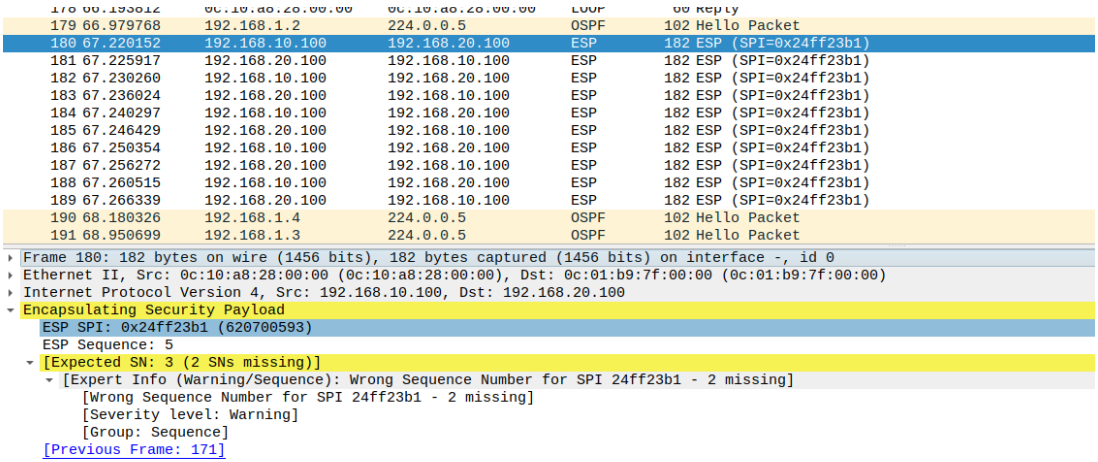


Figure 12: ESP packets captured for PC1 → PC2 ping. All packets carry `SPI=0x24FF23B1`.

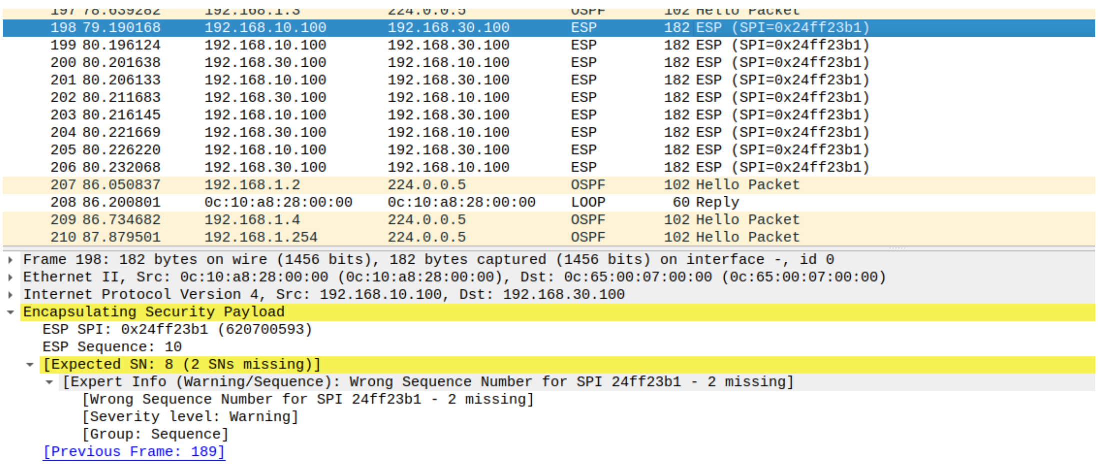


Figure 13: ESP packets captured for PC1 → PC3 ping. Same SPI value despite different destination.

5.2.2 CLI Verification

The `show crypto ipsec sa | include spi` command was run on all three Group Members to confirm the shared SPI at the router level:

```
R2#sh crypto ipsec sa | include spi
     current outbound spi: 0x24FF23B1(620700593)
      spi: 0x24FF23B1(620700593)
      spi: 0x24FF23B1(620700593)
R2#
R3#sh crypto ipsec sa | include spi
     current outbound spi: 0x24FF23B1(620700593)
      spi: 0x24FF23B1(620700593)
      spi: 0x24FF23B1(620700593)
R3#
R4#sh crypto ipsec sa | include spi
     current outbound spi: 0x24FF23B1(620700593)
      spi: 0x24FF23B1(620700593)
      spi: 0x24FF23B1(620700593)
R4#
```

Figure 14: `show crypto ipsec sa | include spi` across all Group Members showing that they all have the same SPI

All three GMs report the same SPI value, confirming they hold an identical TEK. The TEK policy at the Key Server, visible in the `show crypto gdoi ks policy` output (Figure 15), confirms that `spi: 0x24FF23B1` is the single TEK distributed to the group, with `transform: esp-aes esp-sha-hmac` and an original lifetime of 3600 seconds.

```
R1#show crypto gdoi ks policy
Key Server Policy:
For group myGDOIGroup (handle: 2147483650) server 192.168.1.254 (handle: 2147483650):

  # of teks : 1  Seq num : 0
  KEK POLICY (transport type : Unicast)
    spi : 0xF3610BE03FC8CD96B1E069A3815260A3
    management alg     : disabled    encrypt alg       : 3DES
    crypto iv length   : 8           key size          : 24
    orig life(sec)     : 86400       remaining life(sec): 86048
    time to rekey (sec): 85823
    sig hash algorithm : enabled     sig key length    : 162
    sig size           : 128
    sig key name       : myRSAKeys
    acknowledgement    : Cisco

  TEK POLICY (encaps : ENCAPS_TUNNEL)
    spi                : 0x24FF23B1
    access-list        : ICMP
    CKM rekey epoch    : N/A (disabled)
    transform          : esp-aes esp-sha-hmac
    alg key size       : 16            sig key size          : 20
    orig life(sec)     : 3600          remaining life(sec)   : 3249
    tek life(sec)      : 3600          elapsed time(sec)     : 351
    override life (sec): 0             antireplay window size: 64
    time to rekey (sec): 2863


R1#
```

Figure 15: `show crypto gdoi ks policy` on R1 confirming the TEK SPI and policy distributed to all GMs.

5.2.3 Why All SPIs Are Identical

In a traditional point-to-point IPSec VPN, each pair of endpoints negotiates its own independent SA with a unique SPI. In GETVPN, the Key Server generates a single TEK and distributes it to all Group Members simultaneously during the GROUPKEY-PULL exchange. Because every GM holds the same key, any GM can decrypt traffic encrypted by any other GM without prior negotiation. The shared SPI is the identifier for this common key, which is why it appears identically on R2, R3, and R4 and in every ESP packet on the wire, regardless of which pair of hosts is communicating.

It is worth noting that the SPI observed here (`0x24FF23B1`) differs from the one seen during the initial verification (`0x6D2F03B9`). This is because the TEK has a lifetime of 3600 seconds and was refreshed between the two observations. The rekey process — in which the KS distributes a new TEK to all GMs before the old one expires — is examined in the following task.

5.3 TEK Rekey Observation

6 GETVPN Extensions

7 Conclusion